

Cheng-Han Lin

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Three shipped products, each built solo. My work is about making large language models into something that can be trusted: retrieval-augmented generation so the output is traceable, self-hosted inference so sensitive data stays put, and machine learning for the forecasting an LLM is bad at — plus all the backend, mobile and payment infrastructure underneath.

I care about evaluation method, privacy design, and the failure cases most people skip.

Projects

HeartBox — AI mood-journaling platform (RAG + emotion forecasting)

Solo developer · Capstone · heartbox.tw · [source](#)

A mental-health platform built end to end — ML, backend, frontend, mobile and infrastructure. **Live at heartbox.tw, sign in with test1 / test1, no registration.**

- **RAG pipeline** — BGE-M3 embeddings over a ChromaDB store built from seven clinical sources (WHO, APA, NHS, NIMH). Retrieval fires when sentiment falls below -0.4 , returns the top-3 passages, and the model must cite them — so advice is traceable rather than invented.
- **Random Forest forecasting** — 53 features (12 metrics $\times$ 4 lag windows, plus 5 calendar features). 5-fold CV: MAE 0.22 on sentiment and 1.04 on stress (22,796 rows); AUC 0.948 with 88% recall on high-stress days (31,720 rows), deliberately tuned for recall because a missed warning costs more than a false alarm.
- **Model selection** — compared linear regression, a decision tree, Random Forest, XGBoost and an LSTM on data requirements, interpretability, inference cost and overfitting risk. Random Forest was the only one that trains on a few hundred rows, predicts in under 50 ms on CPU, and explains itself.
- **Self-hosted inference** — open-weight LLaVA-v1.6-Mistral-7B for vision and TAIDE-LX-7B, tuned for Traditional Chinese, on my own GPU behind FastAPI and a Cloudflare Tunnel, so journal text never leaves a controlled environment.
- **Privacy and safety** — journals Fernet-encrypted (AES-128-CBC + HMAC-SHA256) with keys held apart from the database, plus 2FA. Three-step separated consent per GDPR Art. 7, where refusing AI-training use costs no functionality, and parental verification for ages 13–17. Crisis-keyword detection surfaces national helplines.

Also delivered: PHQ-9 / GAD-7 assessments, mood trend and correlation visualisations, sleep and habit analytics, a 103-achievement system, and an Android build via Capacitor.

LapseWatch — Subscription renewal reminders

Solo developer · Jun 2026 – present · lapsewatch.smallworks.app

- Notifies **before** a subscription auto-renews, over LINE, email and desktop, with Google Calendar sync and CSV / JSON export.
- A Chrome extension that detects subscriptions while reading **only a narrow window of text around a matched keyword** , to bound what it can see.
- Integrated recurring subscription billing and statutory e-invoicing (ECPay, Taiwan’s e-invoice system).
- Privacy-first: no bank-account linking, no cross-site tracking.

PantryKeeper — Shared household inventory tracker

Solo developer · Jun 2026 – present · pantrykeeper.net

- Multi-user inventory and expiry tracking with per-member permissions, offline support and waste statistics.
- A bulk import parser that accepts free-form data pasted from Notion, Excel or Google Sheets and infers quantity and unit from it.

Skills

Languages	Python, JavaScript, TypeScript, SQL, HTML, CSS
AI / ML	RAG, LangChain, ChromaDB, BGE-M3, self-hosted LLM inference, scikit-learn, Random Forest, feature engineering, cross-validation, model evaluation and selection
Backend	Django, Django REST Framework, Django Channels, FastAPI, Node.js, REST API design, PostgreSQL
Frontend	React, Vite, Tailwind CSS, Recharts, Chrome extensions
Infrastructure	Google Cloud Run, Cloudflare Pages and Tunnel, Capacitor, Git, CI/CD
Security	Fernet / AES encryption, JWT, 2FA, rate limiting, GDPR and Taiwan PDPA consent design
Integrations	Payment gateway and statutory e-invoicing (ECPay, Taiwan), LINE Messaging API, Google Calendar API

Spoken Mandarin Chinese (native), English (professional reading and writing)

Education

**National Chin-Yi University of Technology —
M.S., Computer Science and Information Engineering**

From September 2026 · Taichung, Taiwan

**National Chin-Yi University of Technology —
B.S., Computer Science and Information Engineering**

September 2022 – June 2026 · Taichung, Taiwan

Capstone project: HeartBox (above)

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